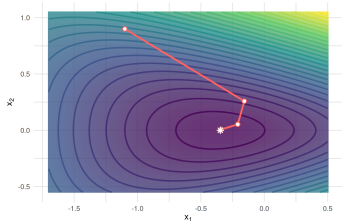


Optimization in Machine Learning

Second order methods

Newton-Raphson vs Gradient Descent



Learning goals

- Comparison of Newton-Raphson and Gradient Descent
- Pure Newton vs relaxed Newton with step size

NEWTON-RAPHSON AND GD (RECAP)

- Gradient Descent: **first order method**
⇒ *Gradient* information, i.e., first derivatives
- Newton-Raphson: **second order method**
⇒ *Hessian* information, i.e., second derivatives

Gradient Descent:

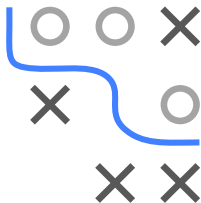
$$\theta^{[t+1]} = \theta^{[t]} - \alpha \nabla f(\theta^{[t]})$$

Pure Newton-Raphson:

$$\theta^{[t+1]} = \theta^{[t]} - (\nabla^2 f(\theta^{[t]}))^{-1} \nabla f(\theta^{[t]})$$

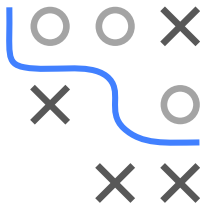
Relaxed/Damped Newton-Raphson:

$$\theta^{[t+1]} = \theta^{[t]} - \alpha (\nabla^2 f(\theta^{[t]}))^{-1} \nabla f(\theta^{[t]})$$



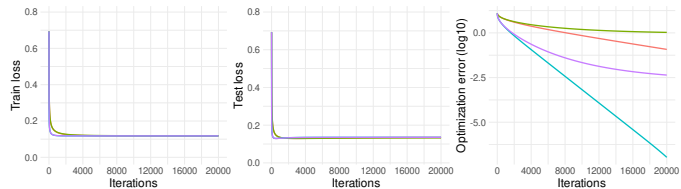
COMPARISON SIMULATION SET-UP

- Comparison of Newton-Raphson, relaxed NR and GD+momentum:
- **Logistic regression** (log loss) simulation with $n = 500$ samples and $p = 11$ features, where $\theta^* = (-5, -4, \dots, 0, \dots, 4, 5)^T$, and $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$ for $\Sigma = \mathbf{I}$ (i.i.d.) or $\Sigma_{i,j} = 0.99^{|i-j|}$ (corr. features)
- To simulate response, we set $y^{(i)} \sim \mathcal{B}(\pi^{(i)})$, $\pi^{(i)} = \frac{1}{1 + e^{-(\mathbf{x}^{(i)})^T \theta^*}}$
- Indep. features result in a condition number of ≈ 2.9 while corr. features yield (moderately) bad condition number ≈ 600
- ERM has unique global minimizer (convexity) but no closed-form solution. We can approximate $\hat{\theta}$ using `glm` solution
- We also track the optimization error $\|\theta - \hat{\theta}\|_2$
- For relaxed NR we use $\alpha = 0.7$ and for GD we set $\alpha = 1$, momentum to 0.8 and use no step size control

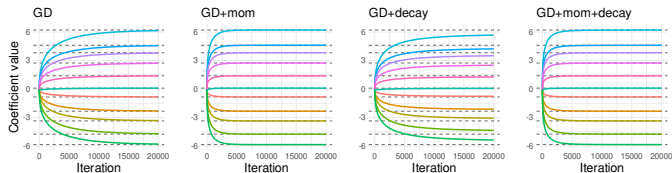


LOGISTIC REGRESSION (GD VARIANTS RECAP)

- Recall comparison of GD variants on log. reg. in last chapter:

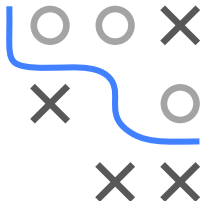


GD — GD+decay — GD+mom — GD+mom+decay Method — GD — GD+decay — GD+mom



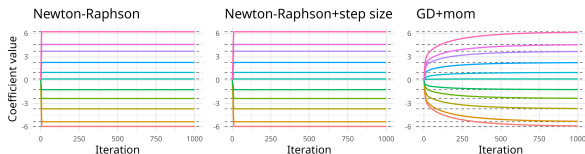
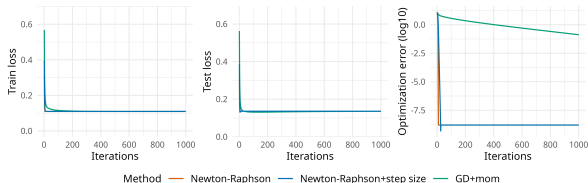
Dotted lines indicate global minimizers.

- GD+momentum** was fastest $\Rightarrow$ now compare w/ Newton-Raphson
- NB:** GD+momentum only converges after several thousand steps



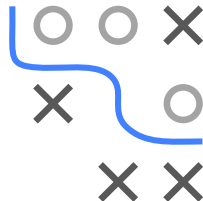
LOGISTIC REGRESSION (GD VS. NR)

- Let's run GD vs. NR for 1000 **steps** (independent features):



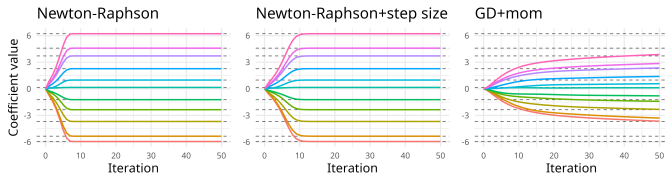
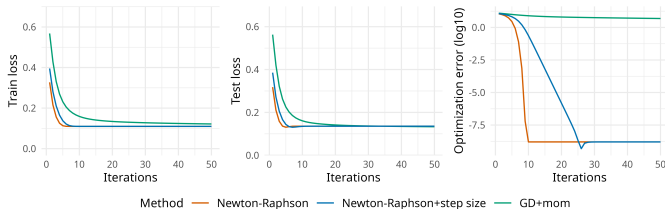
Dotted lines indicate global minimizers.

- NR** and **relaxed NR** $\Rightarrow$ almost instantaneous convergence (see optimization error)
- Using $\alpha < 1$ slightly slows down **relaxed NR**
- GD+mom** several orders of magnitude slower than NR



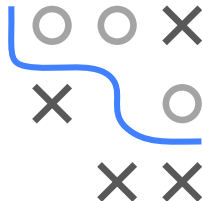
LOGISTIC REGRESSION (GD VS. NR)

- Let's run the same configuration only for 50 **steps** to see clearer picture:



Dotted lines indicate global minimizers.

- NR** takes ≈ 10 steps to reach same optimization error as **GD+mom** after 20,000 steps!
- Relaxed NR** with $\alpha < 1$ shows no advantage here

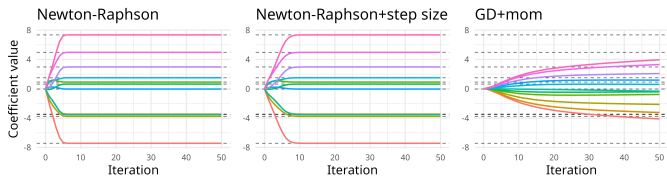
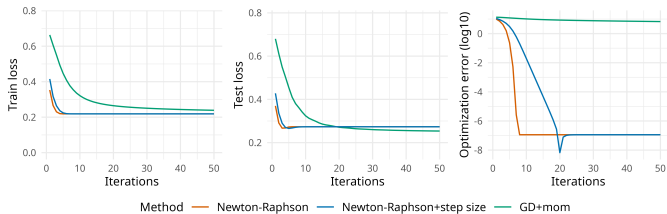


-
- Training loss**
- Y-axis: Training loss (0.1 to 0.5). X-axis: Seconds (0.00 to 0.06).
- Parameter optimization error**
- Y-axis: Optimization error (log10) (-7.5 to 0.0). X-axis: Seconds (0.00 to 0.06).
- Legend: Method — Newton-Raphson (orange) — Newton-Raphson+step size (blue) — GD+mom (green)

- **NR** steps are indeed slower than **GD** steps ($\approx 3\times$ here)
- But each NR step is so much better than GD ($\approx 2000\times$) that per-iteration runtime advantage of GD becomes **irrelevant**

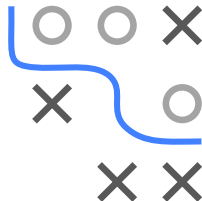
LOGISTIC REGRESSION (CORR.)

- In case of correlated features the results are very similar:



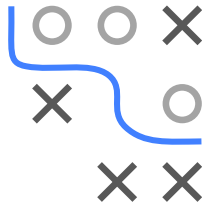
Dotted lines indicate global minimizers.

- **NR's** performance is unaffected by feature correlation
- **GD** iterates become “warped” compared to before



A 3x3 grid with a blue path starting at the top-left cell and ending at the bottom-right cell. The path is composed of three segments: a vertical segment from (1,1) to (2,1), a horizontal segment from (2,1) to (2,2), and a diagonal segment from (2,2) to (3,3). The grid contains the following symbols: (1,1) empty, (1,2) circle, (1,3) cross, (2,1) cross, (2,2) empty, (2,3) circle, (3,1) empty, (3,2) cross, (3,3) cross.

-
- A 3x3 grid with a blue path starting at the top-left cell (row 1, column 1) and ending at the bottom-right cell (row 3, column 3). The path consists of the following cells: (1,1), (1,2), (2,2), (2,3), and (3,3). The cells (1,3), (2,1), and (3,1) are marked with a black 'X'. The cells (1,2), (2,3), and (3,3) are marked with a grey circle.



A 3x3 grid with a blue path starting at the top-left cell (row 1, column 1) and ending at the bottom-right cell (row 3, column 3). The path consists of the following cells: (1,1), (1,2), (2,2), (2,3), and (3,3). The cells (1,3), (2,1), and (3,1) are marked with a black 'X'. The cells (1,2), (2,3), and (3,3) are marked with a grey circle.

-
- A 3x3 grid with a blue path starting at the top-left cell and ending at the bottom-right cell. The path is composed of three segments: a vertical segment from (1,1) to (2,1), a horizontal segment from (2,1) to (2,2), and a diagonal segment from (2,2) to (3,3). The grid contains the following symbols: (1,1) empty, (1,2) circle, (1,3) cross, (2,1) cross, (2,2) empty, (2,3) circle, (3,1) empty, (3,2) cross, (3,3) cross.